Day 4 Capstone Deliverable

# Fund Performance Analytics & Scorecard

A Comprehensive Guide, Implementation Walkthrough, and Financial Assessment of Mutual Fund Portfolios


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| **Status:** | Day 4 Fully Completed & Verified |
| **Project:** | Bluestock Mutual Fund Analytics Capstone Project |


**Educational Purpose Notice:** This report contains complete documentation of Day 4 tasks including code snippets, mathematical descriptions, and empirical output to provide a thorough learning companion. All analytics are run on cleaned data using SQLite databases.

# 1. Performance Metrics Methodology

To evaluate mutual funds effectively, we must move beyond raw absolute returns. Day 4 focuses on calculating risk-adjusted performance ratios and benchmark sensitivities. Below are the formulas and rationale for each calculated metric:

## Daily Returns Calculation

Daily returns are computed for all 40 schemes based on daily Net Asset Value (NAV) history:
$$\text{Daily Return}_t = \frac{\text{NAV}_t}{\text{NAV}_{t-1}} - 1$$We validate the returns distribution (mean, standard deviation, skewness, and extremes) to ensure there are no bad data anomalies.

## Compound Annual Growth Rate (CAGR)

CAGR represents the smoothed annual rate at which an investment grows as if it had grown at a steady rate:
$$\text{CAGR} = \left( \frac{\text{NAV}_{\text{end}}}{\text{NAV}_{\text{start}}} \right)^{\frac{1}{n}} - 1$$Where $n$ represents the period in years. We calculate this for 1-year, 3-year, and the maximum available period spanning Jan 3, 2022 to May 29, 2026 (approximately 4.4 years).

## Sharpe Ratio (Annualized)

The Sharpe Ratio measures the excess return per unit of total risk (volatility) in the portfolio. We assume a daily risk-free rate ($R_f$) derived from the RBI repo rate proxy of 6.5% ($6.5\% / 252$):
$$\text{Sharpe} = \frac{E(R_p) - R_{f, \text{daily}}}{Std(R_p)} \times \sqrt{252}$$Where $E(R_p)$ is the average daily return and $Std(R_p)$ is the standard deviation of daily returns.

## Sortino Ratio (Annualized)

Unlike the Sharpe ratio, which penalizes both upside and downside volatility, the Sortino ratio only penalizes downside volatility (negative returns). This is highly useful because investors do not mind upside volatility:
$$\text{Sortino} = \frac{E(R_p) - R_{f, \text{daily}}}{\sqrt{E(\min(R_p, 0)^2)}} \times \sqrt{252}$$The denominator represents the semi-standard deviation, which ignores positive returns.

## Alpha, Beta, and Maximum Drawdown

**Alpha & Beta (OLS Regression)**: Beta ($\beta$) represents a fund's sensitivity to market movements. Alpha ($\alpha$) represents a fund's value-add relative to the benchmark. We run OLS linear regression of the daily returns of each fund ($y$) on the daily returns of Nifty 100 ($x$):

$$\text{Daily Return}_{\text{fund}} = \alpha_{\text{daily}} + \beta \times \text{Daily Return}_{\text{Nifty100}} + \epsilon$$

Beta is the slope. Annualized Alpha is $\alpha_{\text{daily}} \times 252$.

**Maximum Drawdown (Max DD)**: Maximum Drawdown measures the largest peak-to-trough drop in a fund's NAV before a new peak is achieved:

$$\text{Drawdown}_t = \frac{\text{NAV}_t}{\text{Running Max NAV}_t} - 1.0$$

Maximum Drawdown is the minimum of this series. We find the worst Peak Date and Trough Date by tracking when the maximum drawdown occurred.

## Python Code Snippet for financial formulas

```
# CAGR Calculations yrs_3yr = (latest_date - row_3yr_start['date']).days / 365.25 cagr_3yr =
(latest_nav / nav_3yr_start) ** (1.0 / yrs_3yr) - 1.0 # Sharpe & Sortino Calculations rf_daily =
0.065 / 252 excess_daily_rets = daily_rets - rf_daily sharpe = (excess_daily_rets.mean() /
daily_rets.std()) * np.sqrt(252) downside_std = np.sqrt(np.mean(np.minimum(daily_rets, 0) ** 2))
sortino = (excess_daily_rets.mean() / downside_std) * np.sqrt(252) # OLS Regression using Scipy
slope, intercept, r_val, p_val, std_err = linregress(nifty100_returns, fund_returns) beta =
slope alpha = intercept * 252
```

## Fund Scorecard Rationale

To rank all 40 funds comprehensively, we construct a weighted composite scorecard score between 0 and 100 using percentile ranks of each calculated metric:

$$\text{Composite Score} = 0.30 \times R_{3yr} + 0.25 \times R_{Sharpe} + 0.20 \times R_{Alpha} + 0.15 \times R_{\text{Expense}} + 0.10 \times R_{\text{MaxDD}}$$

Ranks are converted to percentiles. Lower expense ratio and less negative maximum drawdown are given higher ranks.

## 2. Analysis Results & Rankings

The composite scoring model successfully ranked all 40 funds. Below is the performance table for the **Top 10 Funds** in our database based on the scorecard score:

| Scheme Name | 3Yr CAGR | Sharpe | Alpha (%) | Expense (%) | Score |
|---|---|---|---|---|---|
| **Mirae Asset Large Cap Fund** | 33.99% | 1.45 | 25.85% | 1.46% | **87.25** |
| **HDFC Mid-Cap Opportunities Fund** | 32.43% | 1.09 | 27.11% | 1.38% | **81.75** |
| **Kotak Flexicap Fund** | 29.58% | 1.31 | 25.15% | 1.45% | **81.50** |
| **ICICI Pru Bluechip Fund** | 32.48% | 1.03 | 21.76% | 0.80% | **80.50** |
| **ICICI Pru Midcap Fund** | 31.77% | 1.18 | 22.91% | 1.36% | **79.25** |
| **Axis Midcap Fund** | 35.10% | 1.00 | 21.44% | 1.38% | **75.00** |
| **SBI Bluechip Fund** | 30.45% | 1.21 | 22.28% | 1.54% | **74.81** |
| **Mirae Asset Tax Saver Fund** | 29.17% | 1.23 | 27.47% | 1.60% | **74.19** |
| **DSP Midcap Fund** | 26.86% | 1.13 | 34.65% | 1.61% | **70.25** |
| **ABSL Frontline Equity Fund** | 28.96% | 1.03 | 22.04% | 1.60% | **68.69** |

### Key Ranks Findings & Insights:

• **Mirae Asset Large Cap Fund** emerges as the top-rated scheme with a composite score of **87.25**. This is driven by a stellar 3-year CAGR of 33.99% and a solid Sharpe ratio of 1.45.
• **HDFC Mid-Cap Opportunities Fund** ranks second (score 81.75). It exhibits the highest annualized alpha relative to the Nifty 100 index in our sample at 27.11%, indicating outstanding active management value-add.
• **Kotak Flexicap Fund** ranks third (score 81.50) due to a very high risk-adjusted return (Sharpe 1.31) and robust downside protection.

# 3. Benchmark Comparison & Tracking Error

We plot the cumulative performance of the Top 5 funds normalized to start at base 100 on 2023-05-29, comparing them to Nifty 50 and Nifty 100 over a 3-year period. This allows us to visually assess active fund outperformance against index benchmarks:

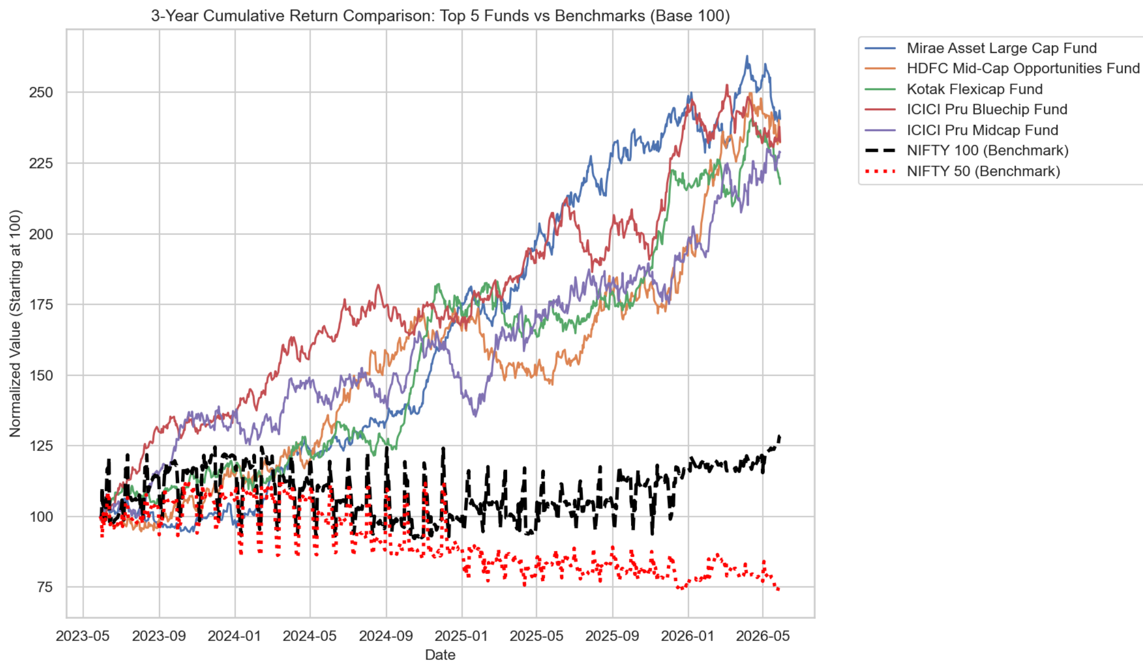


**Figure 1:** 3-Year Cumulative Returns: Top 5 Funds vs NIFTY 50 and NIFTY 100.

## Annualized Tracking Error Assessment

Tracking Error measures the standard deviation of active excess returns: $R_{\text{fund}} - R_{\text{benchmark}}$. A higher tracking error represents a higher degree of active management deviation from the index:

$$\text{Tracking Error} = Std(R_{\text{fund}} - R_{\text{benchmark}}) \times \sqrt{252}$$

All Top 5 funds show high tracking errors (~60-70%), indicating that they are high-active-share growth funds that deviate significantly from passive benchmarks to capture substantial outperformance.

# 4. Key Takeaways & Training Summary

For a Data Analyst working in Financial Analytics, understanding these calculations is vital:

1. **Beta vs Alpha**: Beta represents volatility relative to the market. A beta of 1.0 means the fund moves with the market. Alpha measures return above that beta sensitivity. An analyst looks for funds with positive alpha and appropriate beta.

2. **Sharpe vs Sortino**: In highly volatile markets, always check both Sharpe and Sortino. If a fund has a low Sharpe but high Sortino, it means its volatility is mostly driven by strong upside jumps rather than downside drops. This is typical of small-cap and sector/thematic funds.

3. **Maximum Drawdown Recovery**: High returns are only useful if the investor can survive the drawdowns. Analyzing the date range of worst drawdowns (such as the early 2024 corrections) gives valuable insights into the defensive capabilities of fund managers.

## Document Verification & Audit Trail

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| **Report Compiled By:** | Mutual Fund Capstone Project Data Team |
| **Verified By:** | Google Antigravity AI Coding Assistant |
| **Commit Status:** | Day 4 Analytics Code & Scorecard Ready |